

Bidirectional Transformations with Predicates*

the Dialectica reading

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Abstract

Bidirectional transformations keep two artifacts consistent, restoring consistency when either side changes. A lens is the case where one artifact is a view of the other, consistency is the graph of the read map, and the repairs write whole states rather than edits — so the consistency requirement is a fact about a lens rather than anything the lens carries. A bimorphic lens is a morphism of a dialectica category whose relation has been discarded. Restoring the relation, with values in a lineale rather than in the booleans, lets a lens carry a specification saying where it may be used. Nothing about the lens changes: there is no wrapper, no new datatype and no new way of composing, and a lens either meets a given specification or does not. What the relation supplies is that the specification of a composite is computed from the specifications of its parts. Over a Lawvere quantale the specification becomes a budget, computed by the same formula, which gives a candidate reading of the principle of least change. A third statement is addressed to the optics libraries: copyability turns out to depend only on the update type. This bears on a difficulty the bidirectional-transformations literature has answered differently: lenses built from total functions require *put* to be defined at every source–view pair, and the usual remedies change the type of *put* or place a monad on it, so that the laws must be restated to match. A predicate leaves *put* pure and the laws intact, and records instead where *put* is legitimate.

1 Introduction

This note is about bidirectional transformations, lenses and the Dialectica construction, and the two relations between them are not of the same kind. Lenses are a *special case* of bidirectional transformations — the asymmetric, state-based one — whereas lenses are the Dialectica construction with something *discarded*, namely the relation. The aim here is to put that back. All three are described from the beginning: the audiences for them have very few readers in common, and the lens that sits between the other two is a different object to each, so nothing below is assumed rather than said. Lenses are where we start, since they are the term common to both relations.

A lens says how a change made to a view is carried back into the source it was read from. Bidirectional transformations in general do not compose: Stevens [17] observes that the

*This note grew out of the 2022 Mathematics Research Community on applied category theory, where one of the working groups looked at the relationship between the Dialectica construction and lenses. That group’s own account of it is Capucci, Gavranović, Malik, Rios and Weinberger [4].

consistency relations compose as relations but the composite has no directional maps, since recovering the intermediate artifact would need information neither side retains. Composition is available where the forward map depends on the source alone — which is the lens case, and there the composite is the one the Dialectica construction already had [5].

What the Dialectica construction has in addition is a predicate at each end and a condition saying the propagation respects it. Most lenses are not meant to be used everywhere: the update has to lie in a range, the state has to satisfy an invariant, or the write is simply undefined on some pairs. That restriction is real but it is not part of the lens. It is stated in prose beside the code, established by an argument about the particular lens, and established again about every composite the lens enters. The predicate is where it goes, and because the condition on a composite follows from the two given conditions by transitivity, the restriction is computed rather than re-established.

1.1 Bidirectional transformations

Stevens [17] makes the point that a bidirectional transformation is not primarily a transformation: it is the maintenance of consistency between two artifacts U and V . One posits a consistency relation $R \subseteq U \times V$ together with two directional transformations, each taking both artifacts and returning a repaired version of one of them. Stevens asks three things of them. *Correctness*: the repair does establish consistency. *Hippocraticness*: nothing is done when the two are already consistent. *Undoability*: reverting a change and propagating again returns the other side to what it was. She calls a transformation with all three *coherent*, and shows that very well behaved lenses are coherent in this sense. Only the first two are used below.

We take throughout the *asymmetric, state-based* case: one artifact is a view of the other, the repairs take whole states rather than edits, and R is the graph of *get*. Consistency is then functional, and therefore invisible: it is not a datum that a lens carries but a fact about it. This is worth saying because it locates the predicate of this note. A specification α is *not* a consistency relation. The two run in different directions. Consistency relates the two artifacts to each other, from U to V , and is a relation between the objects; α relates a state to a challenge to it, from U to X , within a single object, and it says how bad that challenge is rather than whether the two sides agree. Bidirectional transformations already have predicates, in the other coordinate.

They also have something closer to α than the consistency relation is, though it is not usually discussed as a predicate. In edit lenses [12] the edits form a monoid acting partially on states, and the domain of definition of that action is exactly a condition on pairs (state, edit). That is α with values in the booleans. What a lineale adds is that the condition grades rather than merely holds or fails; what §2 adds is that it composes. The other way of recording the same partiality is to change the type of *put*, as monadic lenses do [1]; the alternatives are compared in §2.

Well-behavedness in the bidirectional-transformations sense is a set of equations between round-trips — reading after writing returns what was written, and so on. Equations between maps are not conditions of the form $\alpha: U \times X \rightarrow L$, and the morphism condition (4) is an inequality between predicates rather than an identity between maps. So nothing in this note expresses the lens laws, and where a lens is called *conforming* below, that means it satisfies a specification α , not that it satisfies those laws.

1.2 Lenses

A lens is the abstraction of a field of a record. Given a type S of states and a type A of views, it consists of a way of reading the view out and a way of writing an updated view back in:

$$get: S \rightarrow A, \quad put: S \times A \rightarrow S.$$

The first argument of *put* matters: writing a new value back into a record needs the old record, since everything other than the field must be carried along. Lenses compose, and a composite of a lens into a record with a lens into one of its fields is again a lens; this is what makes them useful in practice, where they are the standard way of manipulating nested immutable data.

The version used here is the *bimorphic* one, in which reading and writing may change the types:

$$get: U \rightarrow V, \quad put: U \times Y \rightarrow X, \tag{1}$$

a lens from (U, X) to (V, Y) .¹ Taking $U = X = S$ and $V = Y = A$ recovers the previous paragraph. Composition is

$$(f, F); (g, G) = (g \circ f, (u, z) \mapsto F(u, G(f(u), z))), \tag{2}$$

the update travelling backwards while the state travels forwards. This is the formula Stevens [17] is forced to, in the one case where bidirectional transformations compose at all. So there is no category of bidirectional transformations; there is one as soon as the forward map is restricted the way a lens restricts it, and its composition is the Dialectica composite. Write **Lens**($\mathcal{C}$) for the resulting category, over any cartesian closed $\mathcal{C}$; we take $\mathcal{C}$ to have finite coproducts as well.

Lens laws — that reading after writing returns what was written, and so on — are usually imposed alongside. Nothing below needs them, and, as noted above, nothing below expresses them either.

1.3 The Dialectica construction

Gödel’s Dialectica interpretation [9] assigns to a formula A a formula

$$\exists u \forall x \alpha(u, x)$$

with α quantifier-free: u ranges over *witnesses*, proposed pieces of evidence for A , and x over *counterexamples*, challenges to a proposed witness. The second coordinate goes under three names — counterexamples, challenges, counters — and we use them interchangeably. In the lens vocabulary of §1.2 the two coordinates are the states and the updates: a witness is a state of the source, a counterexample is an update it might be asked to absorb, and α records what is being asked of it when it absorbs that one. A proof of A yields a witness; a proof of an implication yields, in addition, a way of turning a challenge to the conclusion into a challenge to the premiss.

¹The name is Hedges’ [10]. The four type parameters here are not those of a polymorphic type system: everything lives in an ordinary category with finite products. The lenses of the bidirectional-transformations literature are the further special case $U = X$ and $V = Y$, recovered by the next sentence.

The categorical construction takes that shape as its objects. Fix an object L of truth values — a *lineale* [13]: an order with a symmetric monoidal $\otimes$, unit e , and a right adjoint $\multimap$ to it; the two-element boolean algebra is the classical case, with $\otimes = \wedge$. An object of $\text{Dial}_L(\mathcal{C})$ [5, 6] is a triple (U, X, α) with $\alpha: U \times X \rightarrow L$, read as $\exists u \forall x \alpha(u, x)$. A morphism $(U, X, \alpha) \rightarrow (V, Y, \beta)$ is a pair

$$f: U \rightarrow V, \quad F: U \times Y \rightarrow X, \quad (3)$$

subject to the condition

$$\alpha(u, F(u, y)) \sqsubseteq \beta(f(u), y), \quad (4)$$

f carrying witnesses forwards and F carrying challenges backwards. The condition is Gödel’s clause for implication: whatever the challenge y to the conclusion, if u meets the challenge $F(u, y)$ manufactured from it, then $f(u)$ meets y . Reading $\alpha(u, x)$ as “ u meets the challenge x ” is what makes the variance come out this way, and it is the reading the relation had originally.

Two structures on objects will be needed. The product is $A \times B = (U \times V, X + Y, \alpha \& \beta)$, where a challenge is a challenge to one component and $\alpha \& \beta$ asks whichever of the two conditions applies. The tensor is $A \otimes B = (U \times V, X \times Y, \alpha \otimes \beta)$, where a challenge is a challenge to both at once and the two conditions are combined by the $\otimes$ of L . Overloading $\otimes$ for the tensor on objects and the multiplication on L is standard and we keep it; the two are different operations, and which is meant is fixed by what they are applied to.

1.4 Lenses are dialectica morphisms without the relation

Displays (1) and (3) are the same data. A morphism of the Dialectica category is a lens together with a condition, and the compositions agree — (2) is the Dialectica composite. Switching the relation off makes the condition vacuous:

Lemma 1. *For $L = \mathbf{1}$, the one-element lineale, $\text{Dial}_1(\mathcal{C}) \cong \mathbf{Lens}(\mathcal{C})$, with composition agreeing on the nose.*

The containment itself has been remarked on before [11], and on its own it says little: two categories agreeing on objects and morphisms is a fact about shape until something travels along the agreement. What travels is the relation. Lenses are what remains of the Dialectica construction once it is discarded, and the rest of the note is about what it does when it is put back. By Lemma 1 we may speak of a lens and a relation-free dialectica object interchangeably.

2 Where a specification should live

Suppose a lens is not meant to be used everywhere. The update may have to lie in a range, the state may have to satisfy an invariant, or *put* may simply be undefined on some pairs. The condition has to be recorded somewhere, and there are only a few places to put it. It can go into the types, at the cost of needing dependent ones. It can go into the code, as a wrapper that returns an option or an error, at the cost of changing what a lens is and having to say what the laws now mean. Or it can be left where it belongs, as a statement about a lens that is either true or false of it.

The third is what a Dialectica object provides. A specification for a lens with states U and updates X is exactly a map $\alpha: U \times X \rightarrow L$: how bad it is to update state u by x . Condition (4) then says that the lens respects the specifications at its two ends. So a lens does not become a different thing when it acquires a specification. There is no new datatype and no new composition operator; the composite of two lenses is the composite it always was, and the only question is which specifications it respects.

What makes this workable rather than merely tidy is that the question has a best answer, and that the best answers compose. Fix a lens $\ell = (f, F)$ — the pair of (1), abbreviated to a single letter for this section — from (U, X) to (V, Y) , and let L be complete. Given a specification β on the target, ask which specification on the source ℓ could possibly respect. A state u can be challenged by x only through some target challenge y with $F(u, y) = x$, so the demand β makes on (u, x) is whatever all such y demand together:

$$\ell^\# \beta(u, x) = \bigwedge \{ \beta(f(u), y) : F(u, y) = x \}.$$

This is the weakest specification on the source that ℓ carries into β — a precondition, computed from a postcondition and a program. If no y has $F(u, y) = x$ the meet is over the empty set and equals $\top$: an update the lens can never produce at u is unconstrained. In the other direction, a specification α on the source is pushed forward to the strongest one the lens can guarantee downstream,

$$\ell_! \alpha(v, y) = \bigvee \{ \alpha(u, F(u, y)) : f(u) = v \},$$

and the two are adjoint.

Example 2. The composers list of Foster et al. [8] is a source of records (*name, dates, nationality*) and a view of records (*name, dates*), with *get* dropping the nationality. Take U and V to be lists of those records, X and Y their replacements, and $F(u, y)$ the list y with each nationality restored from the record of the same name in u , and set to *unknown* where u has no such record. Let $\beta(v, y) = e$ when y is a well-formed view list and $\perp$ otherwise, so that the target asks nothing beyond well-formedness.

Compute $\ell^\# \beta(u, x)$. A source update x is constrained exactly when $x = F(u, y)$ for some y , and the constraint is then $\beta(\text{get}(u), y)$. If x is reachable from u — if it gives each name either the nationality u already has for it or *unknown* — the meet is taken over the view lists producing it and gives e . If x assigns some name a nationality that u does not have and that is not *unknown*, no y produces it, and the meet is over the empty set and gives $\top$. So the weakest precondition says: this lens may be asked for any update that does not invent a nationality. That is the partiality of *put* in the example, computed rather than stipulated, and recorded as a statement about the lens rather than as a change to its type.

The meet is not a formality here. Many view lists restore to the same source list, since editing a name and editing it back lands where it started, and the obligation at (u, x) is what all of them demand together.

Theorem 3. $\ell_! \dashv \ell^\#$, and the lens ℓ underlies a morphism $(U, X, \alpha) \rightarrow (V, Y, \beta)$ of $\text{Dial}_L(\mathcal{C})$ if and only if $\alpha \sqsubseteq \ell^\# \beta$, equivalently $\ell_! \alpha \sqsubseteq \beta$. Moreover $(\ell' \ell)^\# = \ell^\# \ell'^\#$.²

²The same thing in the language of fibrations, for readers who want it: the forgetful functor $\text{Dial}_L(\mathcal{C}) \rightarrow \text{Lens}(\mathcal{C})$ is a faithful bifibration with posetal fibres. The fibre over a lens is the poset of specifications it respects, the two adjoints above are its reindexings, and faithfulness says that a lens respects a specification in at most one way.

The specification is a precondition, and preconditions compose. This has three readings for a programmer. First, the condition is a Hoare triple for a bidirectional program: $\ell^\# \beta$ is the weakest precondition, and asking whether a lens meets a specification is asking whether its actual precondition is implied by the one you were willing to assume. Second, because the fibres are posets, there is nothing to choose — a lens meets a specification or it does not, and no data is being carried around to record how. Third, $(\ell' \ell)^\# = \ell^\# \ell'^\#$ says that the obligation attaching to a composite is obtained from the obligations of the parts, by composing them in the reverse order. Verification travels with composition instead of starting again at every stage, which is the property a library needs if specifications are to be checked at all.

3 The optics hierarchy

So far one lens at a time, each focusing on a single place in the source. Programs want more, and the lens libraries supply a family — traversals, affine traversals, prisms — differing from lenses only in what the update type is allowed to be, and known collectively as *optics*. Which one may be used where is usually presented as a hierarchy to be learned. It is a question about copying: a traversal reaches many places at once, so the object it focuses on has to be duplicable, and duplication is not free here, $\otimes$ being no product.

In this note an optic is an object of $\text{Dial}_L(\mathcal{C})$, and the familiar names are names for its update type: a *lens* has updates in X , a *traversal* in the free commutative monoid $M(X)$ — the finite multisets on X — and an *affine traversal* in $X + 1$. The optics literature presents these by profunctor classes instead; the comparison in general is not needed here.

Whether an optic may be duplicated is a question about its update type, and about nothing else.

Lemma 4. *A commutative comonoid in $\text{Dial}_L(\mathcal{C})$ is a comonoid in the first coordinate together with a U -indexed family of commutative monoids $(\varnothing_u, \star_u)$ on the second, subject to $\alpha(u, \varnothing_u) \sqsubseteq e$ and $\alpha(u, x \star_u x') \sqsubseteq \alpha(u, x) \otimes \alpha(u, x')$.*

Proof. Weakening $A \rightarrow I$ has forward map $U \rightarrow 1$ and backward map $U \times 1 \rightarrow X$, that is a map $U \rightarrow X$, a point $\varnothing_u$ at each witness. Contraction $A \rightarrow A \otimes A$ has forward map the diagonal and backward map $U \times (X \times X) \rightarrow X$, that is a multiplication $\star_u$ at each witness. The conditions are the displayed ones, and the comonoid laws, which hold separately at each u , are those of $(\varnothing_u, \star_u)$. $\square$

The variance turns the comonoid in the second coordinate into a monoid, and the first coordinate costs nothing, every object of a cartesian category being a comonoid. So:

Theorem 5. *Whether an object is copyable depends only on its update type. An object with updates in X is not copyable; one with updates in $M(X)$ is; one with updates in $X + 1$ is not. So of the three, the traversals are the copyable optics.*

Proof. An optic is uniform in its update type, so the structure Lemma 4 asks of it must be natural in X ; at a single object there is nothing to settle, a monoid on a given X being available by accident. On X itself there is no natural point, so no unit. On $M(X)$ multiset union is a natural commutative monoid. On $X + 1$ naturality leaves a projection or the

constant $*$, unitality forces $(x, *) \mapsto x$ and $(*, y) \mapsto y$, and neither survivor is associative — for the constant, $(x \star y) \star z = z$ while $x \star (y \star z) = x$.³ $\square$

That is the optics hierarchy explained rather than matched: the traversals are the copyable optics, and the reason is that only their update type absorbs two updates into one. The hierarchy is usually a table in a library’s documentation, learned by use; here it falls out of asking which objects carry a commutative comonoid. Nothing beyond the definition of §1 was needed to ask the question.

Remark 6 (order). The traversal of functional programming has *ordered* updates, a list X^* , whereas Lemma 4 asks for a commutative monoid, since contraction is symmetric. So Theorem 5 is a statement about traversals up to order.

4 Quantitative specifications

Until now a specification has only had to be met or not met. Bidirectional transformations usually want more than a verdict: a repair that changes little is better than one that changes a lot, and choosing among the repairs that restore consistency is the principle of least change. A lineale with values in $[0, \infty]$ records that directly, $\alpha(u, x)$ becoming the cost of updating u by x rather than a verdict on it. But a specification then carries two structures, the monoid on the update type of §3 and the $\otimes$ of the lineale, and they are not independent.

Theorem 7. *Let A carry a commutative comonoid whose monoids $\star_u$ are idempotent, and suppose every element of L is a value of α . Then $a \sqsubseteq a \otimes a$ for every a . If L is in addition integral, $e = \top$, then $\otimes$ is idempotent, and an integral commutative idempotent lineale with finite meets has $\otimes = \wedge$, so the specification is purely boolean. A free commutative monoid imposes no such equation, and a traversal may carry any lineale.*

Proof. Idempotence of $\star_u$ in the condition of Lemma 4 at $x' = x$ gives $\alpha(u, x) \sqsubseteq \alpha(u, x) \otimes \alpha(u, x)$, which is the first claim. Integrality gives $a \otimes a \sqsubseteq a \otimes \top = a$, so $\otimes$ is idempotent; then $a \otimes b \sqsubseteq a$ and $a \otimes b \sqsubseteq b$, while $a \wedge b = (a \wedge b) \otimes (a \wedge b) \sqsubseteq a \otimes b$. $\square$

So costs, edit distances and resource counts are available exactly at the optics whose update types are free, and the collapsing cases are confined to plain preconditions. Over the Lawvere quantale [14] $([0, \infty], \geq, +, 0)$, for instance, the condition on a traversal reads as a bound on the total size of the updates it produces, and the bound of a composite is computed by Theorem 3. In the composers example of §2 the same reading turns the precondition into a price: $\sqsubseteq$ is $\geq$, so the meet over the view updates producing a given source update is the largest of their costs, and the obligation on a source update is the worst any view update charges for it.

5 What a predicate buys

The fibrational and profunctorial accounts of optics — lenses as morphisms of fibrations [16], optics as Tambara modules [15, 3], and the treatment of lenses, optics and dialectica

³The two update types are not ad hoc. $M(-)$ is the monad behind the Diller–Nahm variant [7] of the Dialectica interpretation, and $(-) + 1$ is the one Biering studied [2]. Neither is introduced here.

categories as instances of one twisted tower of fibrations [4] — are considerably more general than the setting used here, and they answer questions this note does not ask. The setting here is the narrow one on purpose: an ordinary category with finite products, a fixed pair of types at each end, and a predicate on them. Everything above is a statement about lenses of that shape, which is the shape of the lenses people write.

A specification has so far been attached to a lens from outside: given ℓ and β , one asks whether ℓ meets it. A library wants the other thing too — to speak of the lenses meeting a given specification as a type, so that a conforming lens can be stored, passed to a function, or returned by one. That asks for the collection of them to be an object of the same category. Biering [2] gets close, using Hyland’s comonad on the update type: with updates in $X + 1$ the backward map may decline, returning no update rather than one it cannot justify, and the option of declining is what makes a function space available at all. What she obtains is a weak exponential, a retraction rather than an adjunction, cartesian closure arriving only after the idempotent is split. The gap is the candidates that meet no condition, and closing it asks for an operation on the *first* coordinate: cutting the witnesses down to those satisfying their own specification, which is what subset types are for in the syntactic setting. That operation and the function space it yields are taken up elsewhere.

Three statements have been made, each checkable by unfolding a definition.

Two of them are addressed to bidirectional transformations. Theorem 3 says a specification can be carried without changing the lens, and that the obligation on a composite is computed from the obligations on its parts, so a precondition need not be propagated by hand and need not be re-established at every stage. Theorem 7 says which optics can carry costs: over a Lawvere quantale the specification becomes a budget, and the budget of a composite is computed by the same formula, which is a candidate formal reading of the principle of least change.

The third is addressed to optics. Theorem 5 says copyability is a question about the update type and nothing else, so the lens/traversal/affine hierarchy is a fact about monoids rather than a classification to be memorised.

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